

CONTACT
INFORMATION

Computational Imaging
Biohub SF

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RESEARCH AND
WORK
EXPERIENCE

Computational Imaging - Biohub

Aug 2024 – present

R&D Engineer - Smart microscopy and high-throughput image analysis on Mantis, a high-throughput correlative light-sheet and label-free microscope. Built DynaTrack, a label-agnostic tracking loop that keeps developing zebrafish neuromasts in view for over 70 hours using only bright-field light [1]. Developed the registration, segmentation, and tracking algorithms and curated the infected live-cell dataset behind DynaCLR and DynaCell [2–5]. Core contributor to biohub, the open-source HPC reconstruction library for Mantis data: SLURM-enabled tracking and virtual-staining CLIs, parallel phase reconstruction, registration and stabilization pipelines.

LIDS - Laboratory of Image Data Science

Aug 2021 – Feb 2025

Graduate Researcher, advisor Prof. Alexandre Xavier Falcão - Master's research, in collaboration with the PREDICT-Meso network at the University of Glasgow, on the first fully automated method to detect Early Contrast Enhancement (ECE), a perfusion biomarker of pleural mesothelioma, in 4D DCE-MRI. The existing protocol depended on a mesothelioma specialist placing regions of interest by hand across time points. Designed a pipeline that corrects patient motion with SyN registration, isolates the pleura from a pleural-fluid segmentation, places 4D regions of interest automatically with two complementary supervoxel methods (SLIC and DISF), and detects ECE from their mean-intensity curves, returning both a patient-level classification and a map of ECE-positive regions. Produced a new 4D ground-truth pleural-fluid dataset for 56 patients and showed that SLIC and DISF trade sensitivity for specificity, an observer-independent result validated visually by a clinical specialist. First-author paper at IEEE ISBI 2026 and open-source release [6, 7]; co-authored a superpixel tutorial at SIBGRAPI and the iMig imaging review [8, 9].

PREDICT-Meso Network

Feb 2023 – Jul 2023

Visiting Researcher - Developed an interactive segmentation tool for 4D DCE-MR images of patients suspected of Malignant Pleural Mesothelioma for label ground truth purposes [10].

LIHI - Lab of Image, Intelligence and Innovation

Jan 2020 – Apr 2020

Visiting Student Researcher - Learned to use CNN in TensorFlow for MRI spine segmentation.

LAB2I - Automation and Intelligent Instrumentation Laboratory

Aug 2017-Dec 2019

Undergraduate Research Internship - Developed an embedded system with sensors to monitor the parameters of temperature and electric current of water treatment by electrocoagulation process [11, 12].

Peaply & Co. Data Engineer

May 2021 – Oct 2021

Designing and building systems for collecting, storing, and analyzing data at scale in Pyspark and SQL.

EDUCATION

University of Campinas SP, BR

Aug 2021 – Feb 2025

M.Sc. in Computer Science

- Thesis: Towards Automatic Detection of Pleural Mesothelioma Biomarker in 4D Dynamic MR Imaging [7]
- GPA: 4.0/4.0

State University of Londrina PR, BR

Apr 2016 - Mar 2021

B.Sc. in Electrical Engineering

- Thesis: Study of Deep Learning and its Application to Medical Images

CONFERENCE
PRESENTATION

ISBI 2026 - IEEE International Symposium on Biomedical Imaging Meso-Regions: 4D Segmentation of Early Contrast Enhancement Biomarkers for Pleural Mesothelioma Diagnosis [6].

SPIE BiOS 2026 DynaTrack: label-agnostic smart tracking for mapping dynamics of organ development [1].

iMig 2023 - 16th International Conference of the International Mesothelioma interest group Development of an automated machine learning tool for reporting of MRI-Early Contrast Enhancement in Mesothelioma [3].

SIBGRAPI 2023 - 36th Conference on Graphics, Patterns and Images Workshop - Superpixel segmentation: from theory to applications [8].

ADVANCED
TRAINING

Marine Biological Laboratory (MBL), Woods Hole, MA, USA

2025

Analytical and Quantitative Light Microscopy

COURSEWORK	Graduate Digital Image Processing, Image Analysis, Parallelization and Optimization Algorithms for Machine Learning Graphs, Machine Learning and Pattern Recognition, Deep Learning, Graph Algorithms. Undergraduate Digital Circuits, Analog and Digital Electronics, C programming, Embedded Systems, Digital Signal Processing, Modern Programming Languages, Communication Principles, Electronic Instrumentation, Microelectronics (ARM), Biomedical Engineering, VHDL.	
TEACHING ASSISTANT	MC202 - Data Structure MC832 - Computer Network	Aug 2022 - Nov 2022 Mar 2022 - Jun 2022
VOLUNTEER EXPERIENCE	Techninas - Girls in Tech Student Branch IEEE UEL Women in Engineering IEEE UEL	2018 - 2020 2016 - 2018 2016 - 2018
PUBLICATION	[1] T. M. Theodoro, A. Q. Ryan, R. Banks, E. Hirata-Miyasaki, T. Lao, S. Jaisingh, A. Jacobo, S. B. Mehta, and I. Ivanov, "Dynatrack: label-agnostic smart tracking for mapping dynamics of organ development," in <i>High-Throughput Biophotonics: Imaging, Spectroscopy, and Beyond XI</i>, vol. 13862, pp. 23–31, SPIE, 2026. [2] S. Pradeep, A. Imran, Z. Liu, T. M. Theodoro, E. Hirata-Miyasaki, I. Ivanov, M. Bhave, S. Khadka, H. Woosley, C. Arias, and S. B. Mehta, "Contrastive learning of cell state dynamics in response to perturbations," 2024. [3] E. Hirata-Miyasaki, S. Pradeep, Z. Liu, A. Imran, T. M. Theodoro, I. Ivanov, S. Khadka, M. Bhave, H. Woosley, C. Arias, <i>et al.</i>, "Mapping cell state dynamics with ai-enabled live cell imaging," in <i>High-Throughput Biophotonics: Imaging, Spectroscopy, and Beyond X</i>, p. PC133300R, SPIE, 2025. [4] E. Hirata-Miyasaki, S. Pradeep, Z. Liu, A. Imran, T. M. Theodoro, I. E. Ivanov, S. Khadka, S.-C. Lee, M. E. Grunberg, H. Woosley, M. Bhave, C. Arias, and S. B. Mehta, "Embedding cell state dynamics via contrastive learning of representations of 3d dynamic imaging datasets," in <i>NeurIPS 2025 Workshop for Imageomics: Discovering Biological Knowledge from Images Using AI</i>, 2025. [5] A. A. Kalinin, D. Zheng, T. M. Theodoro, I. E. Ivanov, E. Hirata-Miyasaki, S.-C. Lee, Z. Liu, S. Varra, T. Chandler, S. Pradeep, A. Liu, M. Leonetti, A. Arias, B. Huang, and S. Mehta, "Dynacell: An evaluation framework for dynamic 3D virtual staining of live cells," in <i>The Fortieth Annual Conference on Neural Information Processing Systems Evaluations and Datasets Track</i>, 2026. Poster; accepted. [6] T. M. Theodoro, I. F. Silva, S. Tsim, K. Blyth, and A. X. Falcao, "Meso-regions: 4d segmentation of early contrast enhancement biomarkers for pleural mesothelioma diagnosis," in <i>2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI)</i>, pp. 1–5, 2026. [7] T. M. Theodoro, "Towards automatic detection of pleural mesothelioma biomarker in 4d dynamic mr imaging," master's thesis, University of Campinas (UNICAMP), Institute of Computing, Campinas, Brazil, 2025. [8] I. Borlido, F. Belém, L. Melo, T. M. Theodoro, I. F. Silva, Z. K. G. do Patrocínio Jr, A. X. Falcao, and S. J. F. Guimaraes, "Superpixel segmentation: from theory to applications," in <i>(to appear in) 2023 36th SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI)</i>, 2023. [9] S. G. Armato, S. I. Katz, T. Frauenfelder, G. Jayasekera, A. Catino, K. G. Blyth, T. Theodoro, P. Rousset, K. Nackaerts, and I. Opitz, "Imaging in pleural mesothelioma: A review of the 16th international conference of the international mesothelioma interest group," <i>Lung Cancer</i>, vol. 193, p. 107832, 2024. [10] I. F. da Silva and T. M. Theodoro, "LIDS-UNICAMP: MISE: Multidimensional Image Segmentation Software (initial alpha release)," Aug 2023. [11] T. Theodoro, "Study an Development of an Automation System to Water Treatment by Electrocoagulation Process," in <i>Congress proceedings.</i>, (Londrina), Annual Congress of Scientific Initiation of UEL, ISSN 2447-4118, 2018. [12] T. Theodoro, "Improvement of an Automation System To Water Treatment by Eletrocoagulation Process," in <i>Congress proceedings.</i>, (Londrina), Annual Congress of Scientific Initiation of UEL, ISSN 2447-4118, 2019.	